MILITARY SPECIFICATION

DEMOLITION KIT, TORPEDO BANGALORE, M1A2, HE LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of demolition kit designated as Demolition Kit, Bangalore Torpedo, M1A2, HE.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons: General Specification For.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosive: Sampling, Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8863369 - Demolition Kit, Bangalore Torpedo, M1A2 Assembly

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8863370 - Loading Assembly
9216100 - Box, Packing, Ammunition for Kit, Demolition, Bangalore Torpedo, M1A2

PUBLICATIONS

U.S. ARMY MUNITIONS COMMAND

ET - 8863369 - Equipment Tabulation

(Copies of specification, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The Demolition Kit, Bangalore Torpedo, M1A2 Assembly shall comply with all requirements specified on Drawings (dwgs.) 8863369, 8863370 and with all requirements specified in applicable specifications.

3.3 Functioning.-The loaded bangalore torpedo shall detonate high order when tested as specified in 4.3.2.

3.4 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of explosive on exterior, chips, cracks, dirt, grease, rust, corrosion products and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing the base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificate of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

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Each lot of bangalore torpedo demolition kits shall contain:

a. Parts from not more than one lot interfix number from one manufacturer.

b. Composition A-3 of the same grade from not more than two lots from one manufacturer.

c. Composition B-4 of the same grade from not more than one lot interfix number from one manufacturer which contains TNT from not more than one lot interfix number from one manufacturer.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Booster (see dwg. 9266746 covering a detail of dwg. 8863370).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	True position of detonator well hole with respect to outside diameter	Gage	01001
102.	Diameter of detonator well hole	Gage	01002
103.	Length of end cap and detonator well cavity	Gage	01003

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	01004

4.2.2.2 Assembly, loading, prior to loading and assembly (see dwg. 8863370).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Waterproofness compound missing or inadequate (either end)	Visual	02001
102.	Evidence of foreign matter on threads and detonator well	Visual	02002
103.	Protective coating inadequate on inside	Visual	02003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	02004

4.2.2.3 Assembly, loading, prior to insertion of booster at pouring end (see dwg. 8863370).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Improper depth for end charge, minimum (min.)	Gage	03001
102.	Excess cavitation	Gage	03002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	03003

4.2.2.4 Assembly, loading, prior to assembly of end cap and detonator well assemblies (see dwg. 8863370).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	End charge omitted, improper or inadequate	Visual	04001

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102.	Threads damaged (either end).....	Visual	04002
103.	Evidence of foreign matter on threads or detonator well.....	Visual	04003
104.	Evidence of damage to any component.....	Visual	04004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4).....	Visual	04005

4.2.2.5 Assembly, loading (see dwg. 8863370)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	True position of inside diameter of detonator well with respect to minor diameter of thread (assembled).....	Gage	05001
102.	Total length, maximum (max.).....	Gage	05002
103.	Outside diameter, max.....	Gage	05003
104.	360 degrees curling or crimping omitted, inadequate or improper (either end)..,.....	Visual	05004
105.	End cap seal omitted, inadequate or improper.....	Visual	05005
Minor:	AQL 1.0 percent		
201.	Marking missing, incorrect or illegible.....	Visual	05006
202.	Evidence of poor workmanship (see 3.4).....	Visual	05007

4.2.2.6 Box, packing, prior to sealing (see dwg. 9216100).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Component or assembly missing.....	Visual	06001
102.	Component or assembly incorrectly packed.....	Visual	06002
Minor:	None defined.		

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4.2.2.7 Box, packing, sealed (see dwg. 9216100).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Box, broken or split	Visual	07001
102.	Contents exposed	Visual	07002
103.	Strapping, missing, broken or loose	Visual	07003
104.	Hardware loose, missing or damaged	Visual	07004
105.	Contents loose, when box is tilted	Manual	07005
Minor:	AQL 2.5 percent		
201.	Marking missing, incorrect or illegible	Visual	07006
202.	Strapping improperly assembled	Visual	07007
203.	Hardware improperly engaged	Visual	07008
204.	Car seal omitted, unsealed or improper	Visual	07009
205.	Skids missing or damaged (if applicable)	Visual	07010

4.2.3 Testing

4.2.3.1 Specific gravity of booster (see dwg. 9266746 covering a detail of dwg. 8863370) - Major defect - Code No. 08001.-A single sampling plan shall be used in accordance with MIL-STD-105 using code letter J with an AQL of 0.65 percent. The test shall be performed as specified in 4.3.1 using test equipment specified in 4.2.4.

4.2.3.2 Functioning (see 3.3) - Major defect - Code No. 09001.-Twenty five (25) test assemblies shall be formed from fifty (50) loaded assemblies selected from each lot. A test assembly shall consist of two loaded assemblies attached by a connecting sleeve. If one or more test assembly fails to meet the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.2 using test equipment specified in 4.2.4.

4.2.4 Inspection equipment.-Equipment Tabulation Number ET-8863369 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

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4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3 Test methods and procedures

4.3.1 Specific gravity of the booster.-The booster shall be weighed on a balance and the weight recorded as dry weight. The booster shall then be weighed while completely submerged in water, taking care to remove all air bubbles remaining on the surfaces of the booster. The temperature of the water shall be 68 degrees Fahrenheit (F) plus or minus 2 degrees F. This weight shall be recorded as weight under water. (When submerged, the position of the booster shall be such that no air is entrapped.) Specific gravity shall be calculated as follows:

$$\text{Specific gravity} = \frac{A}{A-B}$$

where:

A = dry weight
B = weight under water

4.3.1.1 Specific gravity of booster, alternate method (mercury displacement).-The specific gravity of the booster shall be determined as specified by method 203.1 of MIL-STD 650.

4.3.2 Functioning.-The test assembly shall be detonated by intiation either by a standard military electric or non-electric blasting cap. Any test assembly which fails to detonate high order shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Level A.-Packaging, packing and marking.-Packaging, packing and marking of demolition kits shall be in accordance with dwg. 9216100.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in the specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection equipment lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Designs" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit code number contained in section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1375-A-129

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